

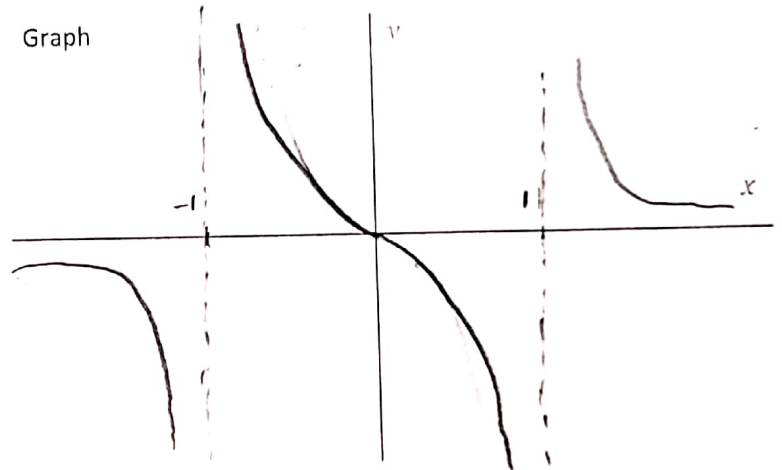
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Fill in the blanks in question 1.

1. Let us consider the rational function $f(x) = \frac{x}{x^2-1}$ for which domain is $R - \{-1, 1\}$ and the root is 0 $\lim_{x \rightarrow (-1)^-} \frac{x}{x^2-1} = \underline{-\infty}$ and $\lim_{x \rightarrow 1^+} \frac{x}{x^2-1} = \underline{\infty}$, hence $x = \pm 1$ are Vertical asymptotes, $\lim_{x \rightarrow -\infty} \frac{x}{x^2-1} = \underline{0}$ and $\lim_{x \rightarrow \infty} \frac{x}{x^2-1} = \underline{0}$, hence $y=0$ is a Horizontal asymptote. $f'(x) = \frac{-x^2-1}{(x^2-1)^2}$ and thus no critical values for relative max or min points. The second derivative is $f''(x) = \frac{2x(x^2+3)}{(x^2-1)^3}$, and hence the only critical value for inflection point is 0.

Table	-1	0	1
f'	-	0	-
f''	-	0	+
f			

Graph

inflection point: (0, 0)

2. Compute the following derivatives.

$(\ln(1 + e^{2x}))'$

$(\arcsin(x^5 + 1))'$

$((\arctan x)^3)'$

$(\cos(\arctan(2x)))'$

$(\sqrt[3]{\arctan(\sec x)})'$

$(\arctan(\sin x))'$

3. Calculate the following limits. (Do not use L'Hospital's Rule)

$\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 3x})$

$\lim_{x \rightarrow 1^-} \frac{3|x-1| + \lfloor x+1 \rfloor + 5x}{\sqrt{2}x} =$

$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2+x+1}}{x-5}$

$\lim_{x \rightarrow 0} \frac{\tan(\sqrt{3}x)}{7x} =$

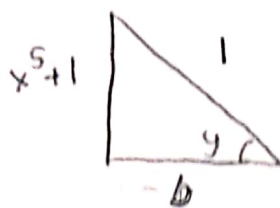
$$\frac{2}{(\arcsin(x^5+1))' = ?}$$

$$y = \arcsin(x^5+1)$$

$$\sin y = x^5+1$$

$$y' \cos y = 5x^4$$

$$y' = \frac{5x^4}{\cos y} \Rightarrow \frac{5x^4}{\sqrt{1-(x^5+1)^2}}$$



$$1^2 = (x^5+1)^2 + b^2$$

$$\sqrt{b^2} = \sqrt{1^2 - (x^5+1)^2}$$

$$b = \sqrt{1-(x^5+1)^2}$$

$$((\arctan x)^3)' = ?$$

$$y = \arctan x$$

$$\tan y = x$$

$$y'(1+\tan^2 y) = 1$$

$$y' = \frac{1}{1+\tan^2 y}$$

$$f(x) = (\arctan x)^3$$

$$f'(x) = \frac{1}{3} (\arctan x)^2 \cdot \frac{1}{1+x^2}$$

$$(\cos(\arctan(2x)))' = ?$$

$$y = \arctan 2x$$

$$\tan y = 2x$$

$$y'(1+\tan^2 y) = 2$$

$$y' = \frac{2}{1+\tan^2 y}$$

$$f(x) = \cos(\arctan(2x))$$

$$f'(x) = -\sin \arctan 2x \cdot \frac{2}{1+4x^2}$$

$$\left(\sqrt[3]{\text{Arctan}(\sec x)} \right)' = ?$$

$$F(x) = \sqrt[3]{\text{Arctan}(\sec x)}$$

$$F'(x) = \frac{1}{3} (\text{Arctan}(\sec x))^{\frac{-2}{3}} \cdot \frac{\sin x}{\cos^2 x + 1}$$

$$(\sec x)' = \tan x \cdot \sec x = \frac{\sin x}{\cos^2 x}$$

$$y = \text{Arctan}(\sec x)$$

$$\tan y = \sec x$$

$$y'(1 + \tan^2 y) = \frac{\sin x}{\cos^2 x}$$

$$y' = \frac{\sin x}{\cos^2 x} \cdot \frac{1}{1 + \sec^2 x}$$

$$y' = \frac{\sin x}{\cancel{\cos^2 x}} \cdot \frac{\cancel{\cos x}}{\cos^2 x + 1}$$

$$y' = \frac{\sin x}{\cos^2 x + 1}$$

$$(\text{Arctan}(\sin x))' = ?$$

$$y = \text{Arctan}(\sin x)$$

$$\tan y = \sin x$$

$$y'(1 + \tan^2 y) = \cos x$$

$$y' = \frac{\cos x}{1 + \tan^2 y} \Rightarrow \frac{\cos x}{1 + \sin^2 x}$$

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$$\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 3x}) \cdot \frac{x - \sqrt{x^2 + 3x}}{x - \sqrt{x^2 + 3x}} = \frac{\cancel{x} - \cancel{x} - 3x}{x - \sqrt{x^2(1 + \frac{3}{x})}} = \frac{-3x}{x - |x|} = \frac{-3x}{2x} = \frac{-3}{2}$$

$$x \rightarrow -\infty \Rightarrow |x| < 0$$

$$\lim_{x \rightarrow 1^-} \frac{3|x-1| + \lfloor x+1 \rfloor + 5x}{\sqrt{2}x} = \frac{3(1-x) + 1 + 5x}{\sqrt{2}x} = \frac{4+2x}{\sqrt{2}x} = \frac{4+2}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

$$x \rightarrow 1^- \Rightarrow |x-1| < 0$$

$$x \rightarrow 1^- \Rightarrow \lfloor x+1 \rfloor = 1$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + x + 1}}{x-5} = \frac{\sqrt{x^2(1 + \frac{1}{x} + \frac{1}{x^2})}}{x(1 - \frac{5}{x})} = \frac{|x|}{x} = \frac{-1}{2}$$

$$x \rightarrow -\infty \Rightarrow |x| < 0$$

$$\lim_{x \rightarrow 0} \frac{\tan(\sqrt{3}x)}{7x} = \frac{\frac{\sqrt{3}}{2} \tan(\sqrt{3}x)}{\frac{\sqrt{3}}{2} \cancel{x}} = \frac{\sqrt{3}}{2}$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$